Task 2 – SQL Foundations Analysis Report

Dataset Overview

- **Database Engine:** MySQL / MySQL Workbench
- **Tables Used:** customers.csv and orders.csv
- **Dataset Scope:** Multi-table relational e-commerce transactions across customer segments, regions, and product categories.

1. SQL Operations & Methods Used

- **Aggregation & Filtering:** Calculated summary totals, averages, and filtered high-value transactions ($\geq \$150.00$) using COUNT, SUM, AVG, WHERE, and ORDER BY.
- **Grouping & Categorization:** Aggregated financial and volume metrics by product line and geographic region using GROUP BY.
- **Relational Joins:** Combined customers and orders tables on customer_id using INNER JOIN to link transaction logs with user demography.
- **Advanced Logic & CTEs:** Applied conditional CASE logic for VIP spend tier segmentation and Common Table Expressions (CTEs) with DENSE_RANK() window functions to isolate top product SKUs.

2. Summary Metrics

- **Total Gross Revenue:** $2,655.00
- **Total Orders Processed:** 12 transactions
- **Total Units Sold:** 21 units
- **Average Order Value (AOV):** $221.25
- **Top Revenue Region:** West Region ($1,020.00)
- **Top Revenue Category:** Electronics ($1,500.00)

3. Key Findings & Analysis Notes

- **Overall Performance:** Generated $2,655.00 in gross sales across 12 transactions, yielding a baseline Average Order Value (AOV) of $221.25.
- **High-Value Orders:** 50% of all orders (6 out of 12) qualify as high-value ($\geq \$150.00$), driven by premium hardware purchases like 4K Monitors ($800) and Standing Desks ($450).

- **Category Share:** Electronics leads overall business performance, generating $1,500.00 (56.5% of gross revenue) across 6 orders, whereas Clothing represents the lowest transaction tier.
- **Regional Contribution:** The West region generates the highest total sales volume ($1,020.00 across 3 transactions), while customer distribution remains balanced across North, South, and West territories.
- **Customer Lifetime Value (LTV):** Accounts are dynamically classified into tiers; VIP accounts (e.g., Priya Nair at $950.00 total spend) generate over 35% of total company revenue.

4. Business Recommendations

- **Inventory Prioritization:** Maintain strong stock levels for Electronics and high-ticket Furniture, as they account for the majority of top-line revenue.
- **Cross-Selling Strategy:** Create bundled offerings for low-performing categories (e.g., Clothing and Office Supplies) to push overall AOV beyond $250.00.

5. File & Folder Organization

The database analysis is structured inside the **Level_1_Beginner/Task_2** directory containing raw CSV sources (customers.csv, orders.csv), the execution script (task2_sql_foundations_analysis.sql), and output validation screenshots (result_screenshot.pdf).